

Technology Stack

Date	29 September 2026
Team ID	SWTID-2026-1156
Project Name	EduGenie: Google Gemini Powered Learning Assistant
Maximum Marks	2 Marks

Define Your Technology Stack:

EduGenie is an AI-powered learning assistant that uses Google Gemini to provide personalized explanations, answers, summaries, study support, and learning guidance. The selected technology stack is designed for a responsive, scalable, secure, and easy-to-maintain application.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side HTML, CSS, JavaScript	HTML5, CSS3, JavaScript	Creates a simple and responsive user interface. JavaScript provides interactive learning features and communicates with the backend.
2	Backend / Server-Side Python / Flask	Python Flask	Handles application logic, user requests, API communication, and secure processing. Flask is lightweight and suitable for an AI learning application.
3	Database / Data Storage User and learning data	MongoDB	Stores user information, learning history, questions, and generated study data using a flexible document-based structure.
4	Cloud / Hosting / Deployment Application deployment	Render	Provides convenient cloud deployment for the Flask application and supports easy hosting and scalability.
5	Version Control & CI/CD Source code management	GitHub	Provides version control, team collaboration, source-code backup, and easy tracking of project changes.
6	Third-Party APIs / Other Tools AI learning service	Google Gemini API	Provides generative AI capabilities for answering questions, explaining concepts, summarizing content, and creating personalized learning support.

Project Summary: EduGenie combines a web-based interface, Flask backend, MongoDB storage, and Google Gemini API to deliver an interactive and personalized learning experience.